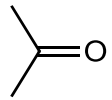
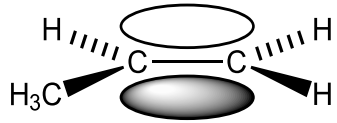
1(a)(i)	(halogens can) remove electron(s) (from a reactant)	**1**

1(b)(i)	**M1** Br—Br is weaker (than C*l*—C*l*) **AND** Br atoms are larger **M2** less (significant) overlap of orbitals (in Br-Br)	**2**

1(b)(v)	**M1** (trigonal) pyramidal **M2** three bonding pairs **AND** one lone pair (of electrons)	**2**

Examiner report missing
9701_w25_qp_24 Question 1

3(a)(i)		**1**

3(a)(iv)		**1**

Examiner report missing
9701_w25_qp_24 Question 3

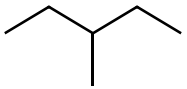
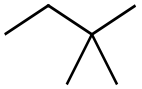
1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(c)(i)	Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons	**1**
1(c)(iv)	**M1** CO_2 has only instantaneous / temporary dipole–induced dipole forces **AND** methanal has permanent dipole–permanent dipole (and id-id) forces **M2** IMF's in methanal are stronger than IMF's in CO_2	**2**

Examiner report missing
9701_w25_qp_23 Question 1

2(a)(i)	delocalised electrons move (through the structure / between layers)	**1**

Examiner report missing
9701_w25_qp_22 Question 2

3(a)	**M1** (family of molecules / compounds / series with) same general formula **OR** (family of molecules where) molecular formula differs (only) by (units of) $CH_{2(n)}$. **M2** similar chemical properties / reactions	**2**

3(d)(i)	**M1**  **M2** 	**2**

3(d)(ii)	**M1** B < A < hexane. **M2** hexane / unbranched has stronger / more IMFs **M3** hexane / unbranched requires most / more energy	**3**

Examiner report missing
9701_w25_qp_22 Question 3

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**

1(c)(i)	Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons	**1**

1(c)(iv)	**M1** CO_2 has only instantaneous / temporary dipole–induced dipole forces **AND** methanal has permanent dipole–permanent dipole (and id-id) forces **M2** IMF's in methanal are stronger than IMF's in CO_2	**2**

Examiner report missing
9701_w25_qp_21 Question 1

2(a)(i)	non-linear	**1**

2(a)(ii)	104.5°	**1**

Examiner report missing
9701_s25_qp_24 Question 2

3(a)(i)	0.00658(0) mol	**1**

3(d)(i)	(electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons	**1**

3(d)(ii)	**M1** HCl is more stable **AND** (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length **M2** (In HCl) stronger attraction between nucleus and shared pair of electrons **OR** (In HCl) (fewer shells of electrons so) less shielding **OR** (In HCl) greater orbital overlap so stronger interaction	**2**

Examiner report missing
9701_s25_qp_24 Question 3

2(a)	Na_2O	**1**

2(c)(i)	SO_3	**1**

2(c)(ii)	• false • Al_2O_3 melting point is lower than MgO • (but) charge on Al ion is greater than that on Mg Any 2 points = 1 mark , all 3 points = 2 marks	**2**

Examiner report missing
9701_s25_qp_23 Question 2

4(a)(i)	Check final graph for range of acceptable answers ● ±(1–0.5) / 100 = ±0.005 ● 0 ● mol dm^{-3} s^{-1} for both Any 2 points = 1 mark, all 3 points = 2 marks	**2**

4(b)(i)	**M1** non-linear **M2** bond angle – 120 > × ⩾ 116°	**2**

4(b)(ii)	**M1** weaker dipole moment (in SO_2) **M2** electronegativity (difference) is larger(/ 1.4) for H and O than for S and O (/ 0.9) **OR** S=O bond is less polar than H—O **OR** stronger dipoles in S=O / SO_2	**2**

Examiner report missing
9701_s25_qp_23 Question 4

1(a)(i)	(both) giant molecular	**1**

1(d)(i)				**3**
	SO_2	–73	pd	
	H_2O	0	H	
	SiO_2	1600	C	
	Na_2O	1132	I	
	MgO	2852	I	

1(d)(ii)	*Evaluate hypotheses based on the information candidates have given in the table* **M1** not enough information **M2** SO_2 and / or H_2O break intermolecular forces **OR** SO_2 and / or H_2O does not break covalent bonds **OR** only SiO_2 breaks covalent bonds	**2**

Examiner report missing
9701_s25_qp_22 Question 1

4(a)(i)	$LiAlH_4$	**1**

4(c)(i)	sp^3	**1**

Examiner report missing
9701_s25_qp_22 Question 4

1(a)(i)	sodium – metallic **AND** sodium oxide – ionic	**1**

1(a)(ii)	delocalised electrons move (through the structure)	**1**

Examiner report missing
9701_s25_qp_21 Question 1

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(b)(i)	green colour (of gas) disappears	**1**
1(b)(ii)	trigonal bipyramid(al)	**1**

Examiner report missing
9701_m25_qp_22 Question 1

2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	**2**

Examiner report missing
9701_m25_qp_22 Question 2

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	**1**

3(d)(i)	instantaneous dipole—induced dipole / id—id / London / dispersion (forces) **AND** permanent dipole—permanent dipole / pd—pd (forces)	**1**

Examiner report missing
9701_m25_qp_22 Question 3

3(a)(i)	<table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table>	**3**
3(a)(iii)	its **delocalised** electrons are free to **move**	**1**
3(d)(i)	**M1** molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	**1**
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces **OR** more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	**1**
3(d)(ii)	tetrahedral	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as MgCl and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiCl_4 is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid SiCl_4 as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiCl_4 molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and SiCl_4 and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	**1**
	M2 $S + O_2 \rightarrow SO_2$	**1**

2(a)(ii)	Al^{3+} 	**1**

2(c)(i)	(molecules) H_2S to H_2Te / they have greater number of electrons	**1**
	stronger instantaneous dipole–induced dipole / London dispersion forces OR more energy required to overcome the instantaneous dipole–induced dipole / London dispersion forces	**1**

2(c)(ii)	(only) H_2O has hydrogen bonding **AND** hydrogen bonding (much) stronger than the other intermolecular forces **OR** H_2O has hydrogen bonding **AND** because of higher electronegativity of O compared to S **OR** stronger van der waals' forces because (only) H_2O has hydrogen bonding **OR** hydrogen bond **AND** increased strength of permanent dipoles in H_2O outweighs the increase in strength of id-id in the others	**1**

Question 2

(a) (i) The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

(ii) Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

(iii) Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

(b) (i) Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of P_4O_{10} and H_2O was not well known, $P(OH)_5$ was a common incorrect answer.

(ii) This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

(iii) This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

(iv) Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of OH^- ions when alcohols dissociate or that water fully dissociates or that alcohols contain OH^- ions.

(c) (i) Many candidate answers referred to H_2O although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

(ii) Identification of hydrogen bonds in H_2O was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to H_2S containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

3(a)(i)	(natural =)lightning	**1**
	(man-made =)internal combustion engines	**1**
3(b)(i)	(+)5 / V	**1**
3(b)(iv)	tetrahedral	**1**
3(f)(i)	**M1** one sigma / σ bond **AND** head-on (overlap of) p / sp (orbitals)	**1**
	M2 two pi / π bond(s) **AND** side-on (overlap of / involving) p (orbitals)	**1**
3(f)(ii)	P≡P is much weaker so P_2 is more reactive (than N_2)	**1**

Question 3

(a) (i) Many candidate answers referred to lightning as responsible for one natural occurrence of NO_2. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

(ii) Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of NO_2 as a catalyst in the production of sulfur-based acid rain.

(iii) Candidates who were familiar with this area of the syllabus described how PAN forms from NO_2. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with NO_2.

(iv) Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of H_2 rather than H_2O.

(v) Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

(b) (i) Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

(ii) The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

(iii) Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

(iv) Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

(c) (i) Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

(ii) A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

(d) Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

(e) Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

(f) (i) This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in P_2 identified as sp^2 or sp^3
- in P_2 overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

(ii) Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

3(a)(i)	<table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table>	**3**
3(a)(iii)	its **delocalised** electrons are free to **move**	**1**
3(d)(i)	**M1** molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	**1**
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces **OR** more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	**1**
3(d)(ii)	tetrahedral	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

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(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

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(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid $SiCl_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid $SiCl_4$ molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and $SiCl_4$ and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

1(a)(i)	**M1** (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond **M2** (N_2 molecules /molecule(s) of N_2) are non-polar	**2**

1(a)(ii)		σ bond	π bond	**4**
	no of bonds	1	2	
	how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)	

Question 1

This question assessed inorganic chemistry, with some aspects of physical chemistry. This tested principally in the AO1 category. Candidates showed some facility of recall but were equally divided in response when asked to apply their knowledge to unfamiliar circumstances.

(a)(i) This standard recall question could be answered with easy reference to syllabus points.

(ii) With reference to the syllabus, orbital overlap can be directly quoted; knowledge of the bonding in N_2 is also straightforward, so the number of σ and π bonds should be readily accessible.

(b)(i) Many candidates answered this question well.

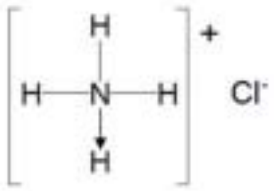
(ii) This question gave rise to many variations of answer. Candidates are advised to be brief with their answers: often, correct responses were contradicted by (incorrect) explanations of valence electrons.

(c)(i) Many candidates answered this question well.

(ii) The reaction of P_4O_{10} was better known than that of Al_2O_3. The latter is not less common but requires better honed balancing skills if constructed rather than written in recall.

(d)(i) Many candidates answered this question well.

(ii) Many candidates were able to identify the type of reaction that occurs.

4(a)	 **OR** Correct representation or label showing **M1** (ionic bond between) NH_4^+ and Cl^- **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 **M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1	**2**

Question 4

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

(a) Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of NH_5 species, with little understanding of the NH_4^+ Cl^- superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

(b)(i) There were many correct answers to this question.

(ii) Different versions of this answer were offered, but a large number were a simple addition of NH_4^+ and OH^-, which did not gain credit.

(c)(i) This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

(ii) This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

(d) This was generally well answered by candidates, but the single-step nature of the S_N2 mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in S_N1. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

(e)(i) This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra H^+ species.

(ii) This question was reasonably well answered.

(iii) This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

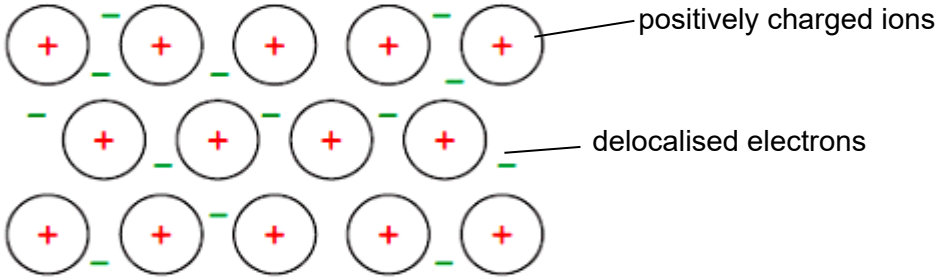
(f) (i) Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

(ii) This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

1(a)		**2**

Mg^{2+}	24	12	10
Al^{3+}	27	13	10

4 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks

1(c)		**1**

labelled diagram shows:

- regular arrangement of minimum of 2 × 2 circles containing + for positive ions
- **AND** surrounded by (sea of) delocalised electrons

Question 1

(a) The majority of answers were correct.

(b) Some excellent explanations were seen. Occasionally the aluminium ion was incorrectly described as larger than the magnesium ion. In general, the explanations focussed, at least in part, on the difference in nuclear attraction which arises as a result of the difference in nuclear charge and the same shielding effect of both species.

(c) Clearly labelled diagrams, which showed a regular arrangement of sodium cations surrounded by delocalised electrons, were seen in a significant proportion of answers. Poorer diagrams showed randomly placed positive ions or made no reference to 'delocalised' electrons. Occasionally, the positive symbols were described incorrectly as 'nuclei' or 'protons' or 'atoms'.

(d) (i) The majority identified the structure and bonding present in silicon. A much smaller proportion used this knowledge to explain why silicon had a high melting point in terms of the high energy requirement required to break many strong covalent bonds. A common misconception described intermolecular forces in a giant molecular structure.

(ii) Of those that annotated Fig.1.1, many correctly compared the melting points of P, S and C*l* with those of the three metals provided. Comparison of the melting points between P, S and C*l* was less well known.

(e) (i) This question proved to be demanding. The state of both sodium oxide and sulfur dioxide under room conditions was not well known. Misunderstanding of the term 'state of oxide' and oxidation number was seen on occasions. A common misconception described aqueous solutions of sodium oxide with a pH of 7.

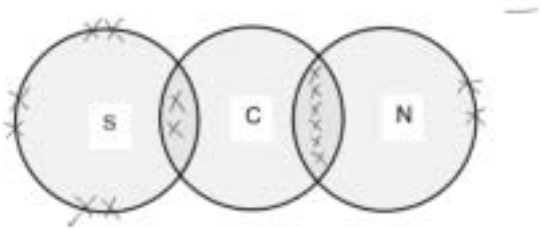
(ii) Correct identification of this reaction as acid–base or neutralisation was common.

(iii) The product for the reaction of P_4O_{10} with an excess of water was well known. Equations were not always balanced.

(f) (i) The majority of answers demonstrated a good understanding of the term amphoteric.

(ii) There was some confusion with this reaction and the reaction of aluminium oxide and sodium hydroxide. The correct formula of the product was not well known.

3(a)	**M1** rates of forward and reverse / backward reactions are equal **M2** no change in measurable properties **OR** concentration of reactants **AND** products remain constant	**2**

3(d)	 **M1** 6 electrons between C and N **AND** 2 electrons between C and S **M2** 3 lone pairs around S **AND** none around C **AND** 1 lone pair around N	**2**

Question 3

(a) Accurate descriptions of this definition were seen. Some weak responses were vague and incomplete, e.g., 'when forward and backward reaction is equal'. The misconception that dynamic equilibrium occurs only when 'concentration of reactants and products are the same' was also seen.

(b) (i) This question proved to be challenging. Many candidates were unable to apply le Chatelier's principle to this novel equilibrium reaction.

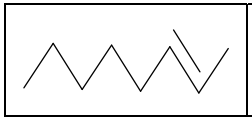
(ii) Some excellent answers, presented clearly and logically, were seen in this demanding question. A significant proportion of candidates confused the terms 'amounts' and 'concentrations'. Understanding of the different mathematical relationship between 2x and x^2 was not always appreciated. Correct deduction of the unit was common.

(c) Many answers were not based on the knowledge that 4s electrons are lost before 3d electrons when the electronic configuration of a transition metal cation is described.

(d) Correct dot-and-cross diagrams were seen. Weaker responses showed correct bonding electron arrangement and ignored the additional electron present in a SCN^- anion.

4(a)	Three-dimensional stereoisomers of $CH_3(CH_2)_5CHBrCH_3$ **M1** correct representation of left-hand answer as 3d structure of one of the enantiomers of $CH_3(CH_2)_5CHBrCH_3$ **M2** correct representation of 3d structure of the second enantiomer of $CH_3(CH_2)_5CHBrCH_3$	**2**

4(d)(i)	elimination	**1**

4(e)(i)	 0 2 6 3 correct answers = 2 marks 2 correct answers = 1 mark 0 or 1 correct answers = 0 marks	**2**

Question 4

(a) Some excellent representations of the three-dimensional structures were seen; these included the approximate bond angles for these tetrahedral structures. Other diagrams showed confusion over the types of symbols used to represent bonds in the plane with those in and out of the plane. In order to remove ambiguity, candidates should be encouraged to use the convention described in the syllabus, under expected conventions for representing organic structures.

(b) Good knowledge of the S_N1 mechanism was described with precision in the best answers. Some confusion regarding the relative charges on the C–Br dipole and either no reference to a lone pair of electrons on O of OH^- or the location of the lone pair of electrons on the H of OH^- were common incorrect details. Some answers confused the S_N2 mechanism details with S_N1.

(c) The observations of these reactions were generally well known. Some answers did not include observations, as directed in the table heading, rather than attempted to predict the product of the reactions.

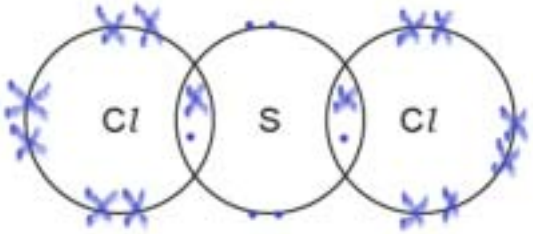
(d) (i) The name of this reaction was quite well known. Addition, substitution, dehydration and cracking were common incorrect answers.

(ii) The reagent NaOH was described in many answers. Confusion of the relevant conditions was seen in weaker responses.

(e) (i) Fully correct answers were seen in around half of responses. Sometimes confusion between two of the types of hybridisations was seen. The weakest answers demonstrated a lack of understanding of the different types of hybridisations which occur in carbon atoms in order to produce single, double and triple covalent bonds.

(ii) It was common for answers to contain features which lead to both types of stereoisomerism rather than focus on the stereoisomerism present in unbranched hydrocarbons. A relatively small proportion of answers showed good understanding of the features of an alkene required in order for it to exist as geometrical (cis/trans) isomers. A lack of precision in answers produced some ambiguity in responses, for example ‘no movement of a double bond’ rather than ‘no rotation of a double bond’.

2(a)(i)	hydrolysis	**1**

2(b)(i)	**M1** 1 shared pair of electrons between each S-Cl shown as •× **M2** 6 non-bonding electrons around each Cl shown as × **AND** 4 non-bonding electrons around S shown as • 	**2**

2(b)(ii)	**M1** shape non-linear **M2** bond angle 103–105(°)	**2**

Question 2

(a) (i) This question was well answered.

(ii) The majority of answers gave a formula for the different Period 3 element when the question asked for a name. Common errors included phosphorus chloride, omitting the oxidation number, and phosphorus pentachloride, which is a solid.

(b) (i) The dot-and-cross diagram of SCl_2 was well answered. A common error included annotating the electrons around the S atom as 'x' in contradiction of the rubric.

(ii) The VSEPR theory was not well known when applied to a central S atom with two bonding pairs and two lone pairs of electrons in the outer shell. The majority of candidates focused on the two bonding pairs in SCl_2, stating that the shape was linear with a bond angle of 180°.

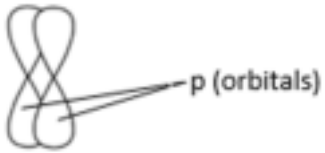
(c) (i) This question was well answered.

(ii) Most candidates gave a correct equation for the reaction of magnesium nitride with water. The most common error was an incorrect formula for magnesium nitride and magnesium hydroxide given as MgOH or MgO.

(iii) The marks were awarded to a minority of candidates. Very few candidates stated that hydroxide ions were present either from the dissociation of $Mg(OH)_2$ in solution, or from a reaction between ammonia and water.

(d) (i) The empirical formula of boron nitride was well answered with a common error of Br used to represent boron.

(ii) Many candidates gave the correct answer of graphite that has a structure similar to that of boron nitride. SiO_2 and graphene were common incorrect answers.

5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	**1**
5(b)(i)	sp^2	**1**
5(b)(ii)	**M1** σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR **M2** π bond - diagram showing how π bond is made from sideways overlap of 2× p– orbitals i.e.  OR  OR  	**2**
5(c)(i)	**EITHER** (pair of) electrons in π bond are further away from the nuclei so weaker attraction **OR** (pair of) electrons in σ bond are closer to the two nuclei so stronger attraction	**1**

Question 5

(a) A minority of candidates gave the correct definition for a covalent bond. The most common errors included omitting a pair of electrons (being shared) and not mentioning that the attraction, for these electrons, was between the nuclei (of the two atoms).

(b) (i) The hybridisation of each carbon, in a C=C bond, was generally well answered.

(ii) Many candidates found this question difficult; some diagrams representing a σ bond were correct showing a linear overlap of two atomic orbitals or hybrids.

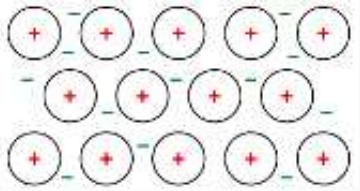
The diagrams for the formation of a π bond were often incorrectly drawn.
The most common mistakes included:

- not labelling p-orbitals, involved in the π bond formation
- drawing only the 'resultant' stage of π bond formation, showing the delocalised electrons above and below the nuclei, but without any labelling.

(c) (i) Credit was rarely awarded and most answers included a statement on how the σ and π bonds are formed by linear and sideways overlap of the relevant orbitals, which was then considered to render the π bond weaker than the σ bond.

The key idea is that during an electrophilic addition reaction the electrons in the π bond are involved as they are further from the carbon nuclei in the C=C bond and thus experience a weaker force of attraction from these nuclei.

(ii) Candidates who understood the reaction usually gained full credit. The main error involved the careless drawing of arrows, which often started from a C atom instead of from the middle of the π bond or from the Br, in a Br^- ion, instead of from a lone pair on the Br^- ion.

1(a)	 labelled diagram shows: • regular arrangement of positive ions • **AND** surrounded by / (sea of) delocalised electrons	**1**

Question 1

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

(a) Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

(b) (i) This question was mostly well answered, though errors of balancing and 'Bi_2' were often seen.

(ii) Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, Bi_2O_3, rather than bismuth itself.

(c) (i) This was answered well, though answers of +6 were commonly seen with respect to bismuth.

(ii) Many different answers were offered here, including $NaBH_4$ and O_2, whereas candidates needed to refer to a compound given in the equation.

(d) (i) This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

(ii) Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

(iii) Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

(iv) Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of CH_3COH is discouraged as a method of representing aldehydes.

(e) The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 dm^3 from 25 cm^3) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

2(a)	<table><tr><td>formula of chloride</td><td>● NaC*l*</td><td>● MgC*l*$_2$</td><td>● SiC*l*$_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**
2(c)(i)	● trigonal bipyramidal ● octahedral	**2**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

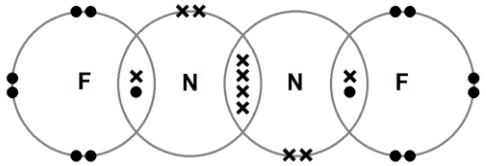
(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

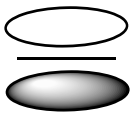
(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

3(a)	strong triple bond / high triple bond enthalpy	**1**
	non-polar (molecule)	**1**

3(d)(i)	 correct bonding pairs	**1**
	lone pairs	**1**

3(d)(ii)	sp^2	**1**

3(d)(iii)	 M1 shows the pi (π) orbital	**1**
	M2 refers to overlap of (two) (2)p orbitals side-on / above and below the plane	**1**

Question 3

This question tested candidates' abilities to apply their knowledge in slightly unfamiliar contexts. In-depth knowledge and recall of nitrogen chemistry was crucial, as was the ability to draw clear, labelled, unambiguous diagrams.

(a) Different types of answer were needed to gain full credit. A strong triple bond and high E_a are in effect making the same point.

(b) (i) There were many correct answers here, with balancing being the main difficulty.

(ii) Very few answers correctly referenced both acidic species, HNO_3 and HNO_2. The latter was rarely seen.

(iii) Candidates deployed a full range of answers referring to environmental issues: specifically, this was looking at photochemical smog.

(c) (i) Candidates found it challenging to apply the definition of standard enthalpy change of formation; attempting to conceive of two numbers that would sum to –46 (or –92), or that combined with the ΔH_f would sum to zero.

(ii) Many candidates were able to give good definitions here.

(iii) Candidates did not perform well on this question. Despite '**rate**' being in bold in the question, the large majority of responses attempted to answer in terms of yield and equilibrium position. Nevertheless, some credit was available for correct and non-contradicted information about a decrease in rate. Where collision theory is referenced, candidates are reminded of the need to discuss the *frequency* (i.e., number *per unit time*) of collisions, rather than simply the number.

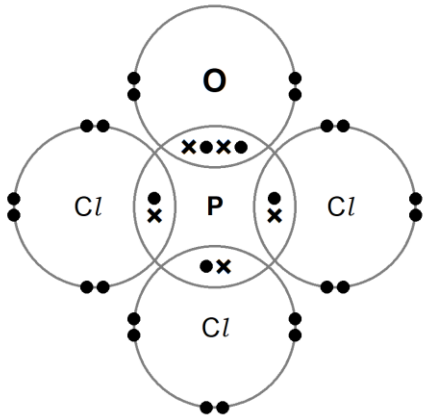
(d) (i) Many diagrams showed a triple bond between the nitrogen atoms or miscounted the number of non-bonding electrons.

(ii) Candidates did not show a good understanding of hybridisation as responses divided reasonably equally between those who offered sp, sp^2 and sp^3 as answers.

(iii) Some good answers were seen here, but many answers could not be awarded full credit. *Sideways* overlap of p orbitals was needed in the response, or a clear and labelled diagram showing this as part of an unambiguous representation of a pi bond.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	**1**
	(vigorous reaction with water suggests) **hydrolysis**	**1**

1(d)(iii)	 Correct bonding e^-	**1**
	rest of electrons **correct (i.e. to make 32 electrons in total)**	**1**

Question 1

(a) (i) Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

(ii) The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

(ii) The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ mol^{-1}, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

(b) (i) The full electronic configuration of the P^{3-} ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

(ii) The decrease in ionic radius from P^{3-} to S^{2-} and Cl^- was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

(c) Observations were required for three tests carried out on separate samples of NaCl(aq).

When Br_2(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added Br_2).

Most answers stated that the observation from the addition of concentrated H_2SO_4 to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

(d) (i) This question provided information that the chemical behaviour of $POCl_3$ is similar to that of PCl_5; $POCl_3$ has a low melting and boiling points and reacts vigorously with water suggesting that the structure and bonding of $POCl_3$ is simple covalent.

A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

(ii) The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for H_3PO_4 and/or incorrect ratio between the water molecules and HCl product.

(iii) Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

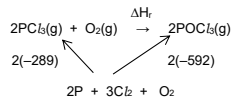
(e) (i) Most candidates were awarded the first mark for the definition of the enthalpy change of formation, for stating 'enthalpy change when 1 mole of a compound is formed'. However, very few correctly stated the second requirement that it is 'from its elements in their standard states'.

Many answers mentioned standard conditions which was not a requirement.

(ii) The calculation of the bond energy of P=O (in PCl_3), using ΔH_f and bond energy (O_2) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:



ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

1(a)(i)	decreases (from Si to S)	**1**

1(b)(i)	total e^- in s subshell	6	6	6	**1**
	total e^- in p subshell	8	9	10	**1**

1(b)(v)	non-linear	**1**

Question 1

(a) (i) The variation in atomic radius from silicon to sulfur was, in general, well known.

(ii) Many candidates appreciated that the differences in melting points were a consequence of the differences in structures. Identification of the types and relative strengths of the forces of attraction broken during melting were referred to less often. Common misconceptions included the presence of intermolecular forces in silicon or did not refer to intermolecular forces, neither in general terms, nor as instantaneous dipole induced dipole attractions in sulfur.

(b) (i) Most candidates correctly calculated the number of electrons in p subshells.

(ii) Many equations showed appropriate species. State symbols, essential in any equation to represent the first ionisation energy process, were included less often.

(iii) Candidates with an excellent knowledge of the trend in ionisation energy across a period identified the correct ionisation energy for phosphorus. They explained both the general trend and the effect of spin pair repulsion on a pair of electrons in a p orbital of sulfur compared to phosphorus. Weaker candidates either identified the value which represents a general trend in ionisation energy across the period and explained this trend or identified the correct value for ionisation energy with details that only explained why sulfur had a lower first ionisation energy than phosphorus.

(iv) This was answered well by most candidates.

(v) Appropriate description of the shape of an SCl_2 molecule as ‘bent’ or ‘v-shape’ was seen often. ‘Linear’ was a common incorrect answer.

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(a)(iii)	dashed (or distinctly labelled) bond between O and H in different molecules of H-O-H and propanoic acid	**1**
	correct sequence of three partial charges over the H-bond $H^{\delta+}$ $O^{\delta-}$—$H^{\delta+}$ OR $O^{\delta-}$ $H^{\delta+}$—$O^{\delta-}$	**1**
	lone pair on O in line with H-bond	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

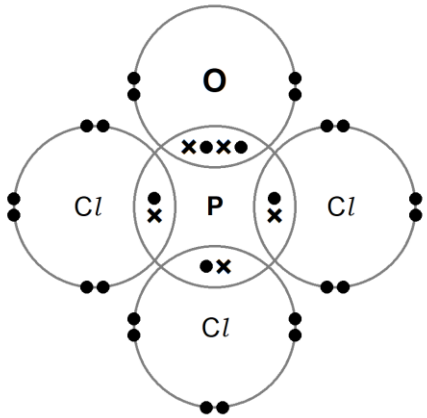
(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	**1**
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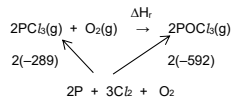
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Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:

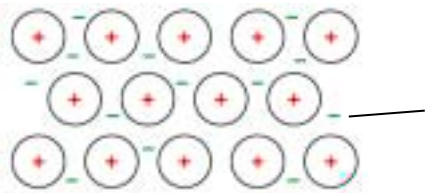


ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

1(a)(i)	giant metallic	**1**

1(a)(ii)	 delocalised electrons **M1** diagram showing minimum of 4 particles • circles containing just a + do not have to be labelled • empty circles / circles with Cu must be labelled + ion / positive ion / cation / Cu^{n+} **AND** with the circles surrounded by electrons shown as e^- / – **OR** little circles labelled delocalised electrons	**1**

Question 1

(a) (i) A high proportion of candidates knew that copper is metallic. Many candidates omitted that the lattice is giant.

(ii) Diagrams showing the metallic bonding as an array of positive metal ions surrounded by delocalised electrons were varied, with many either omitting or not correctly labelling the positive ions and delocalised electrons.

(b) (i) Some candidates incorrectly referred to a mole of ^{12}C atoms or just an atom of ^{12}C.

(ii) Many correct answers were seen. Some candidates omitted the word 'average'.

(iii) The question was quite well answered. The major error included the use of 63 and 65 as the masses of the copper isotopes instead of the data given in the question.

(c) Very few candidates recognised that the ion with an abundance of 23% and an *m/e* of 31.5 was that of $^{63}Cu^{2+}$.

(d) (i) The equation was not well answered overall. Few candidates recognised that it was a redox reaction. Many answers contained electrons on one side of the equation and/or were completely unbalanced.

(ii) The identification of Cu^{2+} as the oxidising reagent in this reaction was generally not well known. Many candidates answered with just 'copper'.

(iii) The full electronic configuration of Cu^{2+} was frequently incorrect with the $4s^2$ electrons still present in many answers.

5(a)	C_3H_7O	**1**

5(c)	type of hybridisation	sp^2	sp^3	**2**
	number of carbon atoms in **X**	2	4	

Question 5

(a) Some candidates deduced the correct empirical formula of C_3H_7O from a molecular formula of $C_6H_{14}O_2$.

(b) (i) Very few candidates used molecular formula, as required in the question, to construct an equation for the reduction reaction.

(ii) Many answers correctly stated that a suitable reducing agent for the reaction is $LiAlH_4$.

(c) Some good answers were seen.

(d) Many candidates correctly stated that **X**, the ketoacid, would react with all three reagents to give effervescence, an orange precipitate and yellow precipitate respectively. Fewer candidates recognised that the diol, **Y**, would only react with the alkaline $I_2(aq)$ as it contained a $CH_3CH(OH)$ moiety.

1(a)(i)	bonding in magnesium – metallic **AND** bonding in sodium chloride – ionic	**1**
1(a)(ii)	bonds in NaC*l* are stronger than bonds in Mg	**1**
1(a)(iii)	**M1** S_8 / molecules of sulfur have more electron**s** (than P_4 / molecules of phosphorus) **M2** S has stronger instantaneous dipole–induced dipole forces (than phosphorus / P)	**2**
1(b)(i)	power of an atom to attract electrons to itself	**1**
1(b)(ii)	(across a period) • increase in nuclear charge • similar shielding • (so) increase in nuclear attraction for bonding / outer / valence electrons **OR** bonding / outer / valence electron(s) are more strongly attracted to nucleus Two correct for one mark, three correct for two marks	**2**
1(b)(iii)	hydrogen bond	**1**
1(b)(iv)	**M1** link shown as a dashed line between the lone pair of electrons from N of one NH_3 to one H on other NH_3 **M2** minimum 3 correct partial charges (on adjacent atoms) over two NH_3 molecules **EITHER** $^{\delta-}$N—$^{\delta+}$H - - - $^{\delta-}$N **OR** $^{\delta+}$H - - -$^{\delta-}$N—$^{\delta+}$H	**2**
1(b)(v)	**M1** O is more electronegative than N **M2** two H-bonds per water molecule : 1 per ammonia molecule.	**2**

Question 1

(a) (i) It was generally well known about the bonding present in both species. Occasionally, candidates incorrectly described the bonding in magnesium as ionic.

(ii) The majority of candidates knew that the higher melting point seen for sodium chloride occurred because of stronger ionic bonds. Some candidates identified that there was a difference in bond strength between the two solids without using the information in the table to identify which had the stronger bond.

(iii) Many candidates were able to explain the difference in melting point in terms of the relative strength of intermolecular forces present. Stronger candidates identified the type of intermolecular forces involved and accounted for the difference in terms of the numbers of electrons present in molecules of sulfur and phosphorus. Some candidates incorrectly explained the difference in melting point being due to the difference in strength of covalent bonds.

(b) (i) This definition was generally well known. Some answers were incomplete or did not identify an atom as the appropriate species involved.

(ii) Many candidates knew that the outer electrons become more strongly attracted to the nucleus. Stronger candidates explained this effect in terms of the change in nuclear charge and the effect of shielding across the period.

(iii) It was generally well known that a hydrogen bond is the strongest intermolecular force present between ammonia molecules.

(iv) Stronger candidates were able to correctly show the role of the lone pair of electrons on the nitrogen atom in the formation of a hydrogen bond between ammonia molecules. Of those, some represented the attraction as a dative covalent bond with an arrow from the lone pair. Identification of appropriate partial charges in diagrams was seen more often. A common misconception showed two hydrogen bonds present between a pair of ammonia molecules to represent a dimer.

(v) The majority of candidates described the increased strength of a hydrogen bond in water and many accounted for the increased strength of the hydrogen bond by describing the difference in electronegativity between atoms of nitrogen and oxygen. A much smaller proportion of candidates also correctly referred to the different number of hydrogen bonds per molecule.

3(a)(i)	KCN in ethanol + heat	**1**

3(a)(ii)	propanenitrile	**1**

3(c)		**2**

type of bond in **X**	number of bonds in **X**
sigma σ	8
pi π	2

Question 3

(a) (i) Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

(ii) The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

(iii) It was generally well known that this type of reaction is a hydrolysis reaction.

(iv) Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

(b) Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

(c) Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.

4(a)	Method 1 • identification of number and type of bonds when all are broken 2O-H + 4C-H + C=C **OR** (2 × 460) + (4 × 410) + 610 • Identification of number and type of bonds when all are made O-H + 5C-H + C-C + C-O **OR** (460 + (5 × 410) + 350 + 360) • calculation of energy to break bonds – energy to make bonds 3170 – 3220 = – 50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks. **OR** Method 2 • identification of number and type of relevant bonds broken (including the new C-H only) 2O-H + C=C **OR** (2 × 460) + 610 • identification of number and type of relevant bonds made (including the new C-H only) O-H + C-H + C-C + C-O **OR** (460 + 410 + 350 + 360) • calculation using energy to break bonds – energy to make bonds 1530 – 1580 = –50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks.	**2**

4(d)		sigma	pi	**2**
	type of orbitals involved in bond	sp^2($-sp^2$)	(2)p(-(2)p)	
	how the orbitals overlap	direct	sideways	
	Two correct for one mark, four correct for two marks.			

Question 4

(a) Stronger candidates showed clear logical working and selection of appropriate data. Common errors included mis-reading the table, using an inappropriate value for a correctly identified bond, or ignoring the C-O bond during the reaction.

(b) Reference to the effect of increasing pressure on the amount of ethanol produced at equilibrium was not always included. Many explanations were given in terms of the change producing a shift to the position of equilibrium to the right; the significance of producing a smaller amount of gas to explain this change was mentioned less frequently.

(c) (i) Many candidates were able to deduce the behaviour of H_3PO_4 as an acid or an electrophile. Some explanations were not specific to step 1 in the reaction.

(ii) Some candidates were able to identify the correct species in step 2. Explanations were often vague, for example, 'seeking electrons' or 'being attacked by a pair of electrons' rather than 'accepting a lone pair'.

(iii) A small proportion of candidates applied knowledge of the drawing of mechanism arrows to represent electron movement when bonds are made and broken in this unfamiliar context. A common error showed arrows pointing in the opposite direction to the way an electron pair moves when a bond is made or broken. Sometimes an arrow did not start from the bond being broken or from the lone pair of electrons on an atom that is used to make a specific bond.

(iv) It was generally well known how a catalyst affects a reaction. Some ambiguity was seen in answers that described the effect of catalysts as increasing the number of collisions between particles rather than increasing the frequency of successful collisions.

(v) Stronger candidates clearly described the regeneration of the catalyst. Some candidates were vague and imprecise, for example 'the catalyst did not change in the reaction'.

(vi) Some candidates showed an excellent understanding of the consequence of more alkyl groups present on the C^+ of the intermediate in terms of a larger inductive effect. Some candidates incorrectly referred to the inductive effect of propan-2-ol rather than the carbocation intermediate which was responsible for the production of propan-2-ol.

(d) The sideways overlap of two p orbitals to form a pi bond between the two carbon atoms was well known. Descriptions of the formation of the sigma bond by direct overlap of sp^2 hybrid orbitals were less common.

1(a)	<table><tr><td></td><td>nucleon number</td><td>number of neutrons</td><td>number of electrons</td></tr><tr><td>Tellurium-130</td><td>130</td><td>78</td><td>52</td></tr></table>	**3**
1(e)	**M1** 6 bonding pairs (of electrons and 0 lone pairs) **OR** coordination number = 6 **OR** $x = 6$ **M2** bonding pairs repel (equally)	**2**

Question 1

(a) This was generally well answered. Some candidates were not familiar with the term nucleon number, often interpreting this as neutron number.

(b) Many candidates answered this question correctly. Common errors included not specifying **1**s and giving 5p as the answer.

(c) Candidates generally answered this well. State symbols were not always included, especially in Te^{+}.

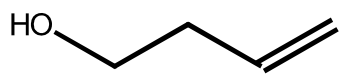
(d) (i) The majority of candidates correctly identified two factors. Many candidates incorrectly stated that the shielding effect remains constant. An 'increase in nuclear charge' was a common incorrect response that negated other descriptions.

(ii) Most candidates recognised a general increase in successive ionisation energies, with a majority of those able to indicate the most significant jump from the 6th to 7th ionisation energy. Many answers incorrectly included a graph with a dip, most often in energy at the $6/7^{th}$ ionisation energy.

(e) Candidates who correctly referenced spin-pair interactions gained credit. Few candidates could use the VSEPR theory correctly.

(f) (i) This was generally not answered well. Many responses included a mixture of numbers and '*x*'s. A small proportion of candidates gave a correctly balanced equation.

(ii) Many candidates correctly identified this as a 'hydrolysis' reaction. A common answer was 'displacement', which was not enough to gain credit.

4(a)(i)		**1**

4(b)(iii)		**2**

type of hybridisation	sp^2	sp^3
number of carbon atoms in **V**	3	1

Question 4

(a) (i) Common errors were to retain the COOH group and to have the C=C double bond reduced.

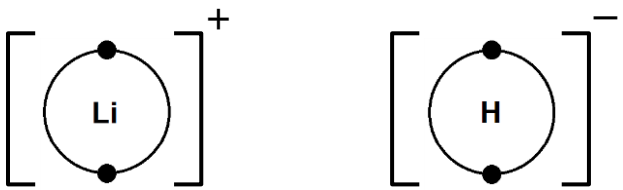
(ii) This was well answered by most candidates.

(b) (i) This was generally well answered, although few candidates stated all the required conditions.

(ii) This was well answered by most candidates.

(iii) A small proportion of candidates gained full credit here.

(c) Many candidates correctly gave pentan-1-ol. A small number of candidates made the link between the yellow precipitate with alkaline I_2(aq) and a $CH_3CH(OH)$– group. Many candidates struggled to synthesise all the detail to present a cogent answer to the question.

1(a)(i)	power of an atom to attract electrons to itself	**1**
1(a)(ii)	• O lower nuclear charge / lower proton number • O has (one) fewer shell than S / less shielding • greater attraction (for nucleus) in O	**2**
1(b)(i)		**2**
1(b)(ii)	non-linear	**1**
1(e)(i)		**2**

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment		✓		✓

Question 1

This question tested ideas of atomic structure, periodicity and some quantitative chemistry. Knowledge of electronegativity was less secure and many Period 3 equations were incorrectly presented as redox reactions.

(a) (i) There were some very good answers here, although many candidates tried to answer in terms of ionisation or electron affinity.

(ii) Few candidates were awarded both marks as this required a balance of points to construct an argument. Spin-pair repulsion arguments were unnecessarily presented on a number of occasions. Many mentioned the distance of the outermost electron from the nucleus without any reference to its significance. Other answers included non-justified references to ionic radius or ionisation energy.

(b) (i) A whole array of different diagrams was seen, with only the ionic form (Li^+ and H^-) acceptable. All electrons, only four in total, were required to be shown.

(ii) Linear, trigonal planar and planar were popular incorrect answers.

(c) (i) The most common error here was the omission of state symbols from the elemental species. Many answers suggested that H_2 was being ionised.

(ii) This was well answered by many candidates.

(iii) Most candidates answered this well, although $3p^6$ was commonly seen.

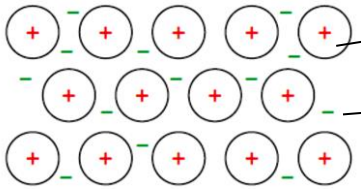
(d) (i) A large number of answers erroneously suggested H_2SO_4 as a product.

(ii) Both Na_2S and Na_2SO_4 appeared regularly, as did H_2. Candidates are advised to consider whether redox has occurred in these reactions.

(iii) Many incorrect versions for the formula of the carbonate were seen such as $Mg(CO_3)_2$ or $MgCO_2$. Many candidates incorrectly gave H_2 as a product from an acid–base reaction.

(e) (i) Many candidates gave the correct overall dipole for S=C=O but then went on to give the remaining three dipoles incorrectly.

(ii) Candidates frequently found it difficult to convert all of the physical quantities to consistent units. Many answers gave the identity as arsenic or some other element and clearly had not used Table 1.2 to identify the compound.

2(a)(i)	 **M1** diagram showing minimum of 4 particles (in total in 2 rows) • circles containing X^{n+} do not have to be labelled •  must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} **AND** with the circles surrounded by electrons shown as e^- / – **OR** little circles labelled electrons	**1**
	M2 label / legend showing **delocalised** electrons	**1**

Question 2

This was an inorganic-chemistry question based around Group 2.

(a) (i) A labelled diagram showing the structure of a metal crystal was required here. Candidates are reminded of the subtle difference between delocalised and free electrons.

(ii) There were many good answers here although some responses continued to use free electrons instead of delocalised.

(b) A large proportion of candidates offered the reaction with steam ($\rightarrow MgO + H_2$) rather than that with cold water, although there were many good answers.

(c) Several answers gave examples of metals which would 'displace' Mg but not Ba. Water was a favourite reagent, with the explanation given as either difference in solubility or a hydrolysis type reaction. Such answers were not sufficiently creditworthy.

(d) (i) This question was answered quite well by candidates.

(ii) A good proportion of candidates understood what was needed, but many spoilt a good answer by stating that X = Ba. The most common error was not to include the stoichiometric factor of ½: such answers tended to have further inconsistency in the subtraction of the A_r of $1\times$ or $2 \times Cl$.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**

3(d)(i)	increases the rate of reaction	**1**
	by providing an alternative reaction pathway of lower E_a	**1**

3(d)(iv)	instantaneous dipole—induced dipole / id—id **AND** permanent dipole—permanent dipole / pd—pd	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.